Cold-Chain Fulfillment Center — Capacity Scaling (unexpected demand increase)

ENG1M-0000001 · Industrial Engineering · Strategy Consulting Hub

Recommendation. Option C — Comprehensive architecture replacement for cold-chain fulfillment center — delivers 13% capacity scaling improvement (target 8.5%) for CAD $676,000 of the CAD $866,000 budget: NPV CAD $2,681,053 at 9.5%, payback 11.1 months.

Capex	NPV @ WACC	IRR	Payback	Benefit/Cost	Run-rate benefit	Levelized cost
CAD $676,000	CAD $2,681,053	119.2%	11.1 months	4.33	CAD $947,537/yr	5.43 CAD per added unit

1. Problem diagnosis

Capacity Scaling sits at 1,190 units/day and must move 8.5% within 3 months under CAD $866,000 (metropolitan public agency; disruption: unexpected demand increase).

- Binding constraint hypothesis: one stage or subsystem limits capacity scaling; confirm with queueing, capacity calculation and bottleneck analysis before committing capital.
- Variability, not average capacity, drives the gap: measure cycle/load variance and its share of lost performance.
- Control and information gap: performance is managed on lagging data, so deviations are corrected late.
- Delivery constraint shaping the solution space: installation restricted to weekends.

2. Options and decision

Opt	Alternative	Capex	Gain	In budget	Meets target	NPV	Payback	Score /10
A	Instrumentation and operating procedure changes for cold-chain fulfillment center	CAD $232,000	4.6%	yes	no	CAD $957,263	10.8 months	4.6
B	Selective retrofit of critical cold-chain fulfillment center subsystems	CAD $502,000	6.9%	yes	no	CAD $1,252,328	14.8 months	5.26
C ✔	Comprehensive architecture replacement for cold-chain fulfillment center	CAD $676,000	13%	yes	yes	CAD $2,681,053	11.1 months	8.55

Criteria and weights: Target attainment 30%, Financial return (NPV) 25%, Payback speed 15%, Downside resilience (-15% perf.) 15%, Budget headroom 15%. Budget is a hard gate.

3. Five-year cash flow (CAD)

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
-CAD $676,000	CAD $747,918	CAD $913,738	CAD $913,738	CAD $913,738	CAD $913,738

4. Scenarios

Scenario	Gain	Capex	NPV	IRR	Payback	Meets target
Bear (-15% perf., +20% capex)	11.05%	CAD $811,200	CAD $1,996,871	81.7%	14.9 months	yes
Base (approved plan)	13%	CAD $676,000	CAD $2,681,053	119.2%	11.1 months	yes
Bull (+15% perf., -10% capex)	14.95%	CAD $608,400	CAD $3,284,656	153.5%	9.1 months	yes

5. Sensitivity (NPV tornado)

Driver	Low NPV	High NPV	Swing
Annual benefit: achieved gain x unit value (+/-20%)	CAD $1,983,686	CAD $3,378,420	CAD $1,394,734
Capital cost (+20% / -20%)	CAD $2,519,897	CAD $2,842,209	CAD $322,313
O&M cost (x2 / x0.5)	CAD $2,551,271	CAD $2,745,944	CAD $194,673
Implementation duration (+50% / -25%)	CAD $2,630,575	CAD $2,731,531	CAD $100,955

6. Implementation roadmap

Horizon	Timing	Workstream	Budget	KPI	Gate
Horizon 1	Month 1	Baseline validation & pilot	CAD $135,200	Pilot reaches >= 2.5% improvement; data quality verified	Go/no-go: pilot result within 20% of model, no safety incidents
Horizon 2	Month 2	Scale-up: Comprehensive architecture replacement for cold-chain fulfillment center	CAD $405,600	>= 6.8% improvement sustained for 4 weeks	Commissioning sign-off and availability at or above constraint
Horizon 3	Month 3	Stabilise, optimise & hand over	CAD $135,200	Full target 8.5% reached; O&M within CAD $33,800/yr	Benefit-tracking dashboard owned by operations

7. Risk register

Risk	L	I	Mitigation	Trigger
Delivered improvement below model	Medium	High	Stage-gated rollout; bear case still yields 11.0% vs 8.5% target	Pilot result < 80% of modelled gain
Capital overrun	Medium	Medium	CAD $190,000 headroom to budget ceiling; fixed-price packages for long-lead items	Committed cost > 90% of H2 allocation
Schedule slip / installation window	Medium	Medium	Pre-fabrication and off-shift installation; pilot proves method before scale-up	Critical-path float < 2 weeks
Safety or compliance non-conformance	Low	Critical	Management-of-change review and HAZOP before each commissioning step	Any open MOC action at gate review
Adoption / operating discipline	Medium	Medium	Operators co-design standard work; KPI dashboard reviewed weekly	Benefit decays > 10% after go-live

8. Judge defence

Q: What changes when unexpected demand increase persists and the selected design underperforms by 15%?

Bear case (performance -15%, capex +20%): 11.0% improvement, NPV CAD $1,996,871, payback 14.9 months. Target is still met.

Q: How does your recommendation change if performance falls 15% and costs rise 20%?

Bear case (performance -15%, capex +20%): 11.0% improvement, NPV CAD $1,996,871, payback 14.9 months. Target is still met.

9. Assumptions

Parameter	Value	Source
Baseline	1,190 units/day	case datum
Target improvement	8.5%	case datum
Capital budget	CAD $866,000	case datum
Timeline	3 months	case datum
Option costs & gains	see options table	case datum
Monetisation	CAD 24.50 contribution/unit x 250 operating days	assumption
Discount rate (WACC)	9.5%	assumption (mid-cap industrial hurdle)
Appraisal horizon	5 years	assumption
Annual O&M	5% of capex	assumption
Benefit during implementation	30% of full run-rate	assumption (pilot ramp)